Antonin Sulc, PhD

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RESEARCH INTERESTS

Neuro-Symbolic AI (Modal Logical Neural Networks, Differentiable Logic) · Belief Functions & Imprecise Probability · Scientific ML (One-Shot PDE Solvers, Physics-Informed Learning) · Agentic AI for Particle Accelerators · Anomaly Detection · Computer Vision (Geometry, Light Fields)

TECHNICAL SKILLS

- **Core AI/ML:** Neuro-Symbolic AI & Differentiable Logic, LLMs & Agentic AI (Fine-tuning, LoRA, RAG), Anomaly Detection, Computer Vision, Uncertainty Quantification (Dempster–Shafer, Evidential Deep Learning).
- **Scientific ML:** Physics-Informed Neural Networks, One-Shot PDE Solvers, Neural SDEs, Geometric Representation Learning.
- **Engineering & MLOps:** Python, PyTorch, Docker, Git, SQL, High Performance Computing (HPC), Linux, Open-Source Packaging (PyPI).
- **Specialized:** Quantum Computing (Qiskit), Geometric Vision, Light Fields.
- **Languages:** English (C1), German (B2), Czech (Native).

ACADEMIC APPOINTMENTS

Innovatrics **Aug 2026 – Present**
Principal Researcher *Prague, Czech Republic*

- **Topic:** Intelligent algorithms for biometrics.

Lawrence Berkeley National Lab (LBNL) **Jan 2025 – Jun 2026**
Researcher (continuing affiliation) *Berkeley, USA*

- **Topic:** Agentic AI for accelerator controls; neuro-symbolic and scientific machine learning.
- Co-developed *Osprey*, a production agentic-AI framework deployed at the Advanced Light Source; its agent plans and runs multistage physics experiments, reducing preparation time about a hundredfold (Physical Review Research, 2026).
- Introduced Modal Logical Neural Networks (differentiable Kripke semantics with learnable accessibility), with applications to multi-agent systems, quantum circuit discovery and finance; released as *torchmodal*.
- Developed *FastLSQ*, a one-shot PDE solver based on Fourier features with exact analytical derivatives.
- Led a multi-laboratory study (Fermilab, Jefferson Lab, LBNL, SLAC) on retrieval-augmented analysis of accelerator electronic logbooks.

Helmholtz Association (DESY/XFEL) **May 2021 – Oct 2024**
Senior Data Scientist *Hamburg, Germany*

- **Topic:** Real-time accelerator control algorithms. **PIs:** Prof. Eichler, Dr. Wilksen.
- Developed unsupervised log anomaly detection and data-driven anomaly detection for SRF cavities at the European XFEL.
- Pioneered domain-specific language models for science: *PACuna* (automated LLM fine-tuning for accelerators) and *ChatQCD* (LLMs for High Energy Physics).

University of Haifa – Marine Imaging Lab **Mar 2020 – Aug 2020**
Visiting Researcher *Haifa, Israel*

- Developed algorithms for robust scene reconstruction and monocular shape from refraction (BMVC 2021, oral).

National Institute of Informatics (NII) **Oct 2018 – Mar 2019**
Visiting Researcher *Tokyo, Japan*

- Research on spectral defocus blur and depth estimation (ICIP 2019, JOSA A 2021).

Vendavo Inc. **Jan 2014 – Dec 2015**
Software Engineer & Data Scientist *Prague, Czech Republic*

- Development of recommender systems for B2B pricing.

EDUCATION

University of Konstanz **2015 – 2020**
PhD in Artificial Intelligence (Computer Vision) *Germany*

- Grade: *Magna Cum Laude* (1.0) · Advisor: Prof. Dr. Bastian Goldluecke
- Focus: Non-standard geometries, light fields, photogrammetry

SOFTWARE & OPEN SOURCE

torchmodal PyPI · GitHub
Differentiable Modal Logic for PyTorch, enabling the training of Modal Logical Neural Networks (MLNNs).

torchbelief PyPI · GitHub
Differentiable belief functions and imprecise probabilities for PyTorch: fuses conflicting sources (Dempster, Yager, TBM, Dubois–Prade, PCR, cautious rules), blames the faulty one, and abstains when evidence is inconclusive.

osprey-framework PyPI · GitHub
A scalable, production-ready framework for orchestrating agentic AI in safety-critical facility operations. Deployed at the Advanced Light Source.

FastLSQ PyPI · GitHub
Solves linear and nonlinear PDEs in one shot via Fourier features with exact analytical derivatives, bypassing automatic differentiation and iterative training.

event2vector PyPI · GitHub
A scikit-learn style library for learning composable, interpretable representations of discrete event sequences. Published in PMLR Vol. 282.

enn-public GitHub
Deep evidential networks with differentiable belief-function layers; mass on the ignorance set acts as an "I do not know" signal.

PACuna GitHub
Automated data collection and fine-tuning pipelines adapting LLMs to particle-accelerator domains.

PUBLICATIONS

1. **A. Sulc**. Neural Networks with Differentiable Belief-Function Layers. In *Belief Functions: Theory and Applications*, Springer, pp. 61–71, 2027.
2. **A. Sulc**. Continuous Modal Logical Neural Networks: Modal Reasoning via Stochastic Accessibility. *arXiv:2603.04019*, 2026.
3. **A. Sulc**. Modal Logical Neural Networks for Financial AI. *ICLR Workshop on Financial AI (FinAI)*, 2026.

4. **A. Sulc**. Differentiable Modal Logic for Multi-Agent Diagnosis, Orchestration and Communication. *3rd International Conference on Neuro-Symbolic Systems (NeuS)*, 2026. **Oral.**

5. **A. Sulc**. FastLSQ: Solving PDEs in One Shot via Fourier Features with Exact Analytical Derivatives. *ICLR Workshop on AI & PDE*, 2026.

6. **A. Sulc**. Differentiable Logical Programming for Quantum Circuit Discovery and Optimization. *arXiv:2602.08880*, 2026.

7. T. Hellert, D. Bertwistle, S. C. Leemann, **A. Sulc**, M. Venturini. Agentic artificial intelligence for multistage physics experiments at a large-scale user facility particle accelerator. *Physical Review Research* 8, L012017, 2026.

8. T. Hellert, J. Montenegro, **A. Sulc**. Osprey: Production-ready agentic AI for safety-critical control systems. *APL Machine Learning* 4, 016103, 2026.

9. **A. Sulc**. Event2Vec: A Geometric Approach to Learning Composable Representations of Event Sequences. *Proceedings of Machine Learning Research (PMLR)* 282:492–512 (NeurIPS NeurReps Workshop), 2026.

10. **A. Sulc**. Modal Logic Neural Networks. *20th International Conference on Neurosymbolic Learning and Reasoning (NeSy)*, 2026. **Oral.**

11. **A. Sulc**. Quantum Noise Tomography with Physics-Informed Neural Networks. *NeurIPS ML4Physics Workshop*, 2025.

12. **A. Sulc**, T. Hellert, A. Reed, A. Carpenter et al. eLog Analysis for Accelerators: Status and Future Outlook. *16th International Particle Accelerator Conference (IPAC)*, 2025.

13. **A. Sulc**, P. L. S. Connor. QCD in Language Models: What do they really know about QCD? *EPS-High Energy Physics*, 2025.

14. **A. Sulc**, P. L. S. Connor. ChatQCD: Let Large Language Models Explore QCD. *42nd International Conference on High Energy Physics (ICHEP)*, 2024.

15. **A. Sulc**, T. Hellert, R. Kammering, H. Hoschouer, J. St. John. Towards Agentic AI on Particle Accelerators. *NeurIPS ML4Physics Workshop*, 2024.

16. **A. Sulc**, G. Hartmann, J. Maldonado, V. Kain et al. Towards Unlocking Insights from Logbooks Using AI. *15th International Particle Accelerator Conference (IPAC)*, 2024.

17. **A. Sulc**, R. Kammering, A. Eichler, T. Wilksen. PACuna: Automated Fine-Tuning of Language Models for Particle Accelerators. *NeurIPS ML4Physics Workshop*, 2023.

18. **A. Sulc**, A. Eichler, T. Wilksen. Unsupervised Log Anomaly Detection with Few Unique Tokens. *IET Information Security*, 2023.

19. **A. Sulc**, A. Eichler, T. Wilksen. A data-driven anomaly detection on SRF cavities at the European XFEL. *Journal of Physics: Conference Series* 2420, 012070, 2023.

20. **A. Sulc**, I. Sato, B. Goldluecke, T. Treibitz. Towards Monocular Shape from Refraction. *British Machine Vision Conference (BMVC)*, 2021. **Oral.**

21. S. Ishihara, **A. Sulc**, I. Sato. Depth Estimation Using Spectrally Varying Defocus Blur. *Journal of the Optical Society of America A* 38(8), 2021.

22. S. Ishihara, **A. Sulc**, I. Sato. Depth from Spectral Defocus Blur. *IEEE International Conference on Image Processing (ICIP)*, 2019.

23. O. Johannsen, **A. Sulc**, B. Goldluecke. What Sparse Light Field Coding Reveals About Scene Structure. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

TALKS & MEDIA

- **Oral talk:** Modal Logic Neural Networks. NeSy 2026, Lisbon.
- **Oral talk:** Differentiable Modal Logic for Multi-Agent Diagnosis, Orchestration and Communication. NeuS 2026, University of Southern California.
- **Workshop talks:** FastLSQ (ICLR 2026 AI & PDE); MLNNs for Financial AI (ICLR 2026 FinAI).
- **Oral talk:** Towards Monocular Shape from Refraction. BMVC 2021.
- **Media:** Agentic AI at the Advanced Light Source covered by the NVIDIA Blog and Berkeley Lab ATAP; quantum circuit discovery featured by Quantum Zeitgeist and Berkeley Lab press; interview "3Q4 with Antonin Sulc" (ATAP, LBNL).

SERVICE & LEADERSHIP

- **Program Chair:** Foundations of Agentic Systems Theory (FAST) @ AAAI 2025.
- **Scientific Chair:** ICFA Beam Dynamics Mini-Workshop on ML for Particle Accelerators.
- **Reviewer:** CVPR, NeurIPS (2023–2025), BMVC, ICLR.